Lecture 5: Causality, Experimental and Quasi-Experimental Methods

Live Session

Purpose of this live session

- **Follow up** on math derivations on
  - The exact percentage change in the log-level functional form.
- **Recap** the pre-recorded lectures
- **Test** your understanding of the pre-recorded lectures
- **Follow up** on the application of Difference in Differences (DiD) estimation:
  - Policy application: School construction in Indonesia
- **Practice** with random assignment

Follow up on math derivations

Logarithmic functional forms : the log-level functional form

- We have already seen the following **log-level** functional form:

$$\widehat{\ln w} = \hat{\beta}_0 + \hat{\beta}_1 math + \hat{\beta}_2 edu \tag{27}$$

- Looking at the predicted change:

$$\Delta\widehat{\ln w} = \hat{\beta}_1 \Delta math + \hat{\beta}_2 \Delta educ \Rightarrow \hat{\beta}_1 = \frac{\Delta\widehat{\ln w}}{\Delta math} \tag{28}$$

- There is a simple calculation available to compute the exact percentage change

$$\%\Delta y = 100.[exp(\beta_1) - 1] \tag{29}$$

(where $\%\Delta y$ means "percent change")

- For small changes in $\ln y$, it is a mathematical fact that $\%\Delta y \approx 100.\Delta \ln(y)$ so:

$$\hat{\beta}_1 \times 100 \approx \frac{\%\Delta w}{\Delta math} \tag{30}$$

- For large values of changes in $\ln w$, the approximation is not very accurate

Recap of pre-recorded lecture videos

Outline

- Causal effect
- Omitted variable bias
- Experimental evidence
  - Policy application: High-quality pre-school education
- Quasi-experimental evidence
  - Policy application: School construction in Indonesia

Causal Effect

- Suppose we estimate the following regression in the data:

$$\ln w_i = \beta_0 + \beta_1 D_i + \epsilon_i \tag{4}$$

- β_1 is the average effect of going to university

$$\beta_1 = E(\ln w_i | D_i = 1) - E(\ln w_i | D_i = 0) \tag{5}$$

- Denote $W_i = \ln w_i$, then:

$$\beta_1 = E(W_{1,i} | D_i = 1) - E(W_{0,i} | D_i = 0) \tag{6}$$

- β_1 compares the average log wages of those who went to university with the average log wages of those who didn't

$\rightarrow$ **Does β_1 measure the causal effect of education?**

Omitted variable bias

- More able people could get more schooling and at the same time earn more not because of the additional schooling but just because they are more able

$$\ln w_i = \beta_0 + \beta_1 D_i + \beta_2 A_i + \epsilon_i \tag{7}$$

where A = ability (e.g. as measured by IQ)

- If we do not observe A and therefore estimate:

$$\ln w_i = \beta_0 + \beta_1 D_i + \epsilon_i \tag{8}$$

- We have an **omitted variable bias** problem (violation of (A-2))
  - In this case, we talk about **ability bias**

Omitted variable bias

- Recall the true model:

$$\ln w_i = \beta_0 + \beta_1 D_i + \beta_2 A_i + \epsilon_i \tag{9}$$

  - We assume that $E(\epsilon_i | D, A) = 0$
  - In other words, once we *control* for education and ability, (A-2) would hold
- This requires having measure of ability, which is far from always being the case!

Omitted variable bias

- Suppose we don't have a measure of ability, but the true model is equation (9)
- Expected log wages of individuals who went to university

$$\begin{aligned} E(\ln w_i|D_i=1) &= E(\beta_0+\beta_1\times 1+\beta_2 A_i+\epsilon_i|D_i=1) \qquad (10)\\ &= \beta_0+\beta_1+\beta_2 E(A_i|D_i=1)+\underbrace{E(\epsilon_i|D_1=1)}_{=0} \end{aligned}$$

- Expected log wages of individual who didn't go to university

$$\begin{aligned} E(\ln w_i|D_i=0) &= E(\beta_0+\beta_1\times 0+\beta_2 A_i+\epsilon_i|D_i=0) \qquad (11)\\ &= \beta_0+\beta_2 E(A_i|D_i=0)+\underbrace{E(\epsilon_i|D_i=0)}_{=0} \end{aligned}$$

- So the difference between the two is:

$$\begin{aligned} & E(\ln w_i|D_i=1)-E(\ln w_i|D_i=0) \qquad (12)\\ = & \Big(\beta_0+\beta_1+\beta_2 E(A_i|D_i=1)\Big)-\Big(\beta_0+\beta_2 E(A_i|D_i=0)\Big) \qquad (13)\\ = & \beta_1+\beta_2\Big(E(A_i|D_i=1)-E(A_i|D_i=0)\Big) \qquad (14) \end{aligned}$$

Omitted variable bias

$$E(\ln w_i|D_i = 1) - E(\ln w_i|D_i = 0) = \beta_1 + \beta_2\Big(E(A_i|D_i = 1) - E(A_i|D_i = 0)\Big)$$

- The last term $E(A_i|D_i = 1) - E(A_i|D_i = 0)$ is the average ability differential between individuals who chose to go to university and those who chose not to
- It is likely that ability and education are positively correlated, that is:

$$E(A_i|D_i = 1) > E(A_i|D_i = 0) \tag{15}$$

- And that ability positively influences wages, so $\beta_2 > 0$.
- If we cannot control for ability differentials between individuals, then the OLS coefficient we estimate will effectively be:

$$\hat{\beta}_1 = \beta_1 + \underbrace{\beta_2\Big(E(A_i|D_i = 1) - E(A_i|D_i = 0)\Big)}_{\text{omitted ability bias } (>0)} > \beta_1 \tag{16}$$

Some solutions to the omitted variable bias problem

- However we can never be sure that we're controlling for all omitted factors
- Better solution is to find a source of *exogenous* variation in the treatment
  - A variable that causes individuals' education to change in a way that is uncorrelated with their unobservable characteristics

The experimental ideal: Random assignment solves the selection problem

- If we could **randomly** assign people to go to university or not, we could compare their earnings and infer from this comparison an unbiased estimate of the return to education
- Random assignment of D_i (treatment) makes treatment independent of potential outcomes W_i
- Because we randomly choose the value of D for everyone in the sample, we can be pretty sure that D is uncorrelated with any of the individuals' characteristics (including ability)

$$\begin{aligned}
\hat{\beta}_1 &= E(\ln w_i | D_i = 1) - E(\ln w_i | D_i = 0) \\
&= \beta_1 + \underbrace{\beta_2 \Big(E(A_i | D_i = 1) - E(A_i | D_i = 0) \Big)}_{=0 \quad \text{because Corr(D,A) = 0}} \\
&= \beta_1
\end{aligned}$$

- Random assignment of the sort would allow us to get an unbiased estimate of the returns to a university degree

Remarks on random assignments

- Experiments are becoming more and more common in economics
- They allow researchers to cleanly estimate the causal effect of a "treatment"
- However:
  - Raise ethical issues
  - Expensive to implement
  - Hard to generalize results out of the sample so the results possibly have little external validity
- What should we do in absence of an experiment?

Quasi-experiments

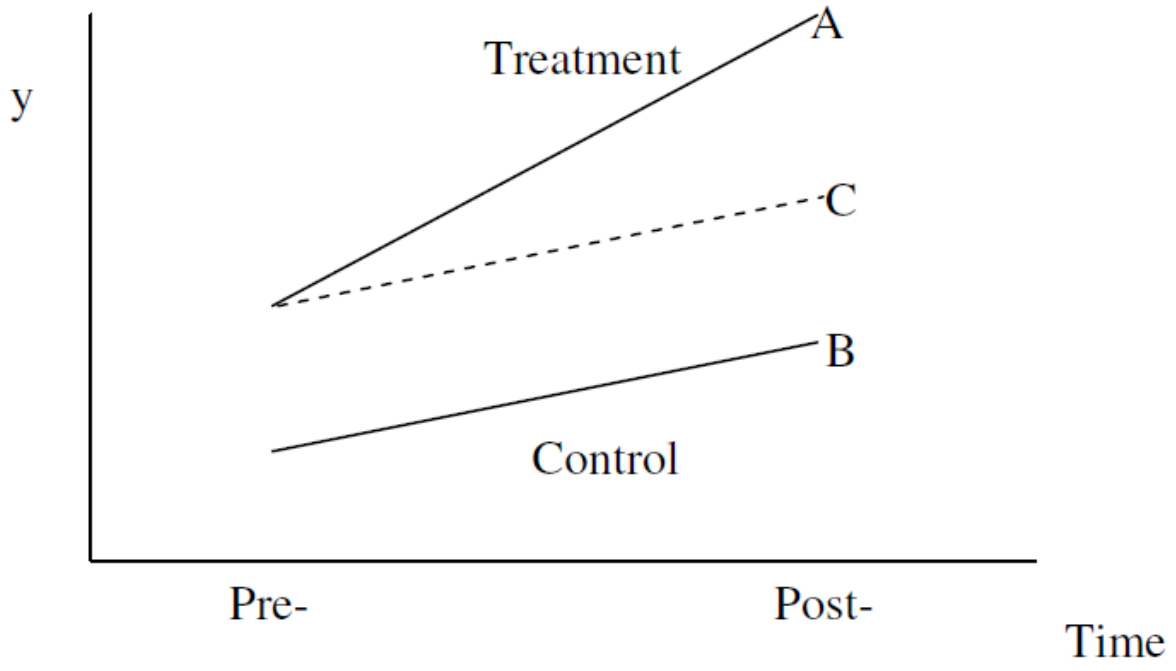
- The idea of is to try to mimic an experiment

→ Try to find equivalents of a 'treatment' and 'control' groups in which everything apart from the variable of interest (or other things that can be controlled for) are assumed to be the same

- This is often a difficult claim to make, as it is rarely possible to do this perfectly
  - Observed differences between treatment and control groups may be the result of some other omitted factors
- An easier assumption to make is that, in the absence of treatment, the unobserved differences between treatment and control groups are the same over time

Differences-in-differences

- In this case, one can use pre-treatment data to estimate the normal difference between treatment and control group and then compare this with the post-treatment difference



Differences-in-differences

- If one just used data from the post-treatment period, then one would estimate the treatment effect as the distance AB
  - This estimate is based on the assumption that the only reason for observing a difference in outcome between treatment and control is the receipt of the treatment
- Difference-in-Difference (DD or DiD) estimator will take the normal difference between control and treatment as CB and estimate the treatment as AC
  - This is valid insofar as the trend in y is the same in both treatment and control

Differences-in-differences

- Notation
  - μ_{it}: mean of the outcome in group i at time t
  - $i = 0$ for the control group, $i = 1$ for the treatment group
  - $t = 0$ for pre-treatment period, $t = 1$ for the post-treatment period
- The idea behind the DD estimator can be illustrated using a two-by-two table:

	Treatment	Control	Difference
Post-treatment	μ_{11}	μ_{01}	$\mu_{11} - \mu_{01}$
Pre-treatment	μ_{10}	μ_{00}	$\mu_{10} - \mu_{00}$
Difference	$\mu_{11} - \mu_{10}$	$\mu_{01} - \mu_{00}$	$(\mu_{11} - \mu_{01}) - (\mu_{10} - \mu_{00})$

- DD estimator $= (\mu_{11} - \mu_{01}) - (\mu_{10} - \mu_{00}) = (\mu_{11} - \mu_{10}) - (\mu_{01} - \mu_{00})$

How can one estimate this in practice?

- Assume we have data on outcomes before and after the treatment for both groups
- One way is to write the DD estimator is to estimate the following regressions:

$$y_{it} = \beta_0 + \beta_1 T_i + \beta_2 Post_t + \beta_3 T_i \times Post_t + \epsilon_{i,t} \tag{17}$$

  where
  - T_i= 1 if the individual is in the treatment group, =0 if in the control group
  - $Post_t$ = 1 in the post-treatment period and 0 in the pre-treatment period
- The DD estimator is the OLS estimate of β_3, the coefficient on the interaction between T_i and $Post_t$.

Test your understanding of the pre-recorded lectures

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Multiple Choice

Omitted variable bias occurs when which of the following conditions are satisfied:

- When the omitted variable is correlated with an included independent variable and has an effect on the dependent variable
- When the omitted variable is correlated with an included independent variable but has no effect on the dependent variable
- When the omitted variable is correlated with the endogenous regressor but uncorrelated with the error term

Multiple Choice

Which of the following statements is true?

- Random assignment provides an exogenous source of variation in the treatment.
- Random assignment requires that all individuals respond identically to the same policy.
- Random assignment works by eliminating individual differences across treatment and control groups

Multiple Choice

Which of the following statements correctly represents the key identifying assumption of the difference in differences estimation?

- In the absence of the treatment, the difference between the treatment and control group would remain the same over time
- Random assignment in a sample that is large enough ensures that differences in individual level characteristics cancel out on average.
- The two groups are the same in terms of observable and unobservable characteristics.

Follow up on the application of Difference in Differences (DiD) estimation

Policy application: School construction in Indonesia

- The INPRES program (1974-1978):
  - Oil crisis creates large windfall for Indonesia
  - Suharto uses oil money to fund school construction
  - Close to 62,000 schools got built by national government
  - More schools built in areas which started with less

The return to education in Indonesia

- Strategy: *difference-in-difference estimation*
  - Data on children born before and after program (pre vs. post)
  - Data on children born in communities where many schools were built (treatment), those where few schools were built (control)
  - Difference-in-difference estimate of program impact compares pre vs. post differences in treatment vs. control communities
- Intuitively, difference-indifference estimation asks:
  - *After controlling for time trends and unchanging differences between treatment and control communities, do children who were born into areas with more newly built INPRES schools get more education and higher earnings?*

The return to education in Indonesia

- In practice, the DD estimator is:

$$\begin{aligned}\text{diff-in-diff} &= \text{post-pre estimator}_{\text{treatment}} - \text{post-pre estimator}_{\text{control}} \\ &= \text{T-vs-C estimator}_{\text{post-treatment}} - \text{T-vs-C estimator}_{\text{pre-treatment}}\end{aligned}$$

The return to education in Indonesia

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	Dependent variable: years of schooling		
	Many schools built	Few schools built	Difference
Under 7 in 1974	8.49	9.76	-1.27
Over 11 in 1974	8.02	9.40	-1.38
Difference	0.47	0.36	0.11

- Younger children (reached school age after INPRES) in areas where the program built a large number of schools are the treatment group

The return to education in Indonesia

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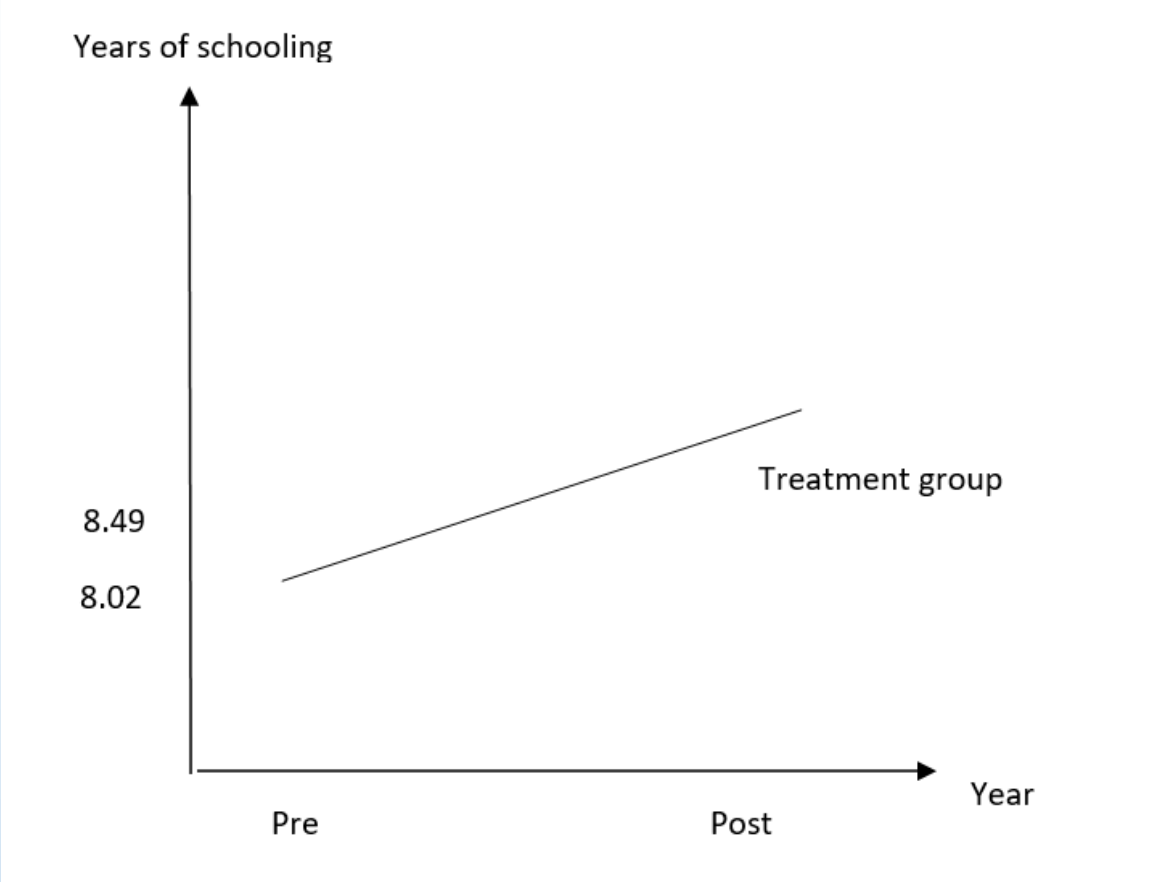
Short Answer

What's the problem with pre-post anlaysis?

Pre-Post analysis



The return to education in Indonesia

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- Pre vs. post analysis does not control for time trends
- Indonesia is getting wealthier over time, so younger children (those entering school after the program) may get more education anyway

The return to education in Indonesia

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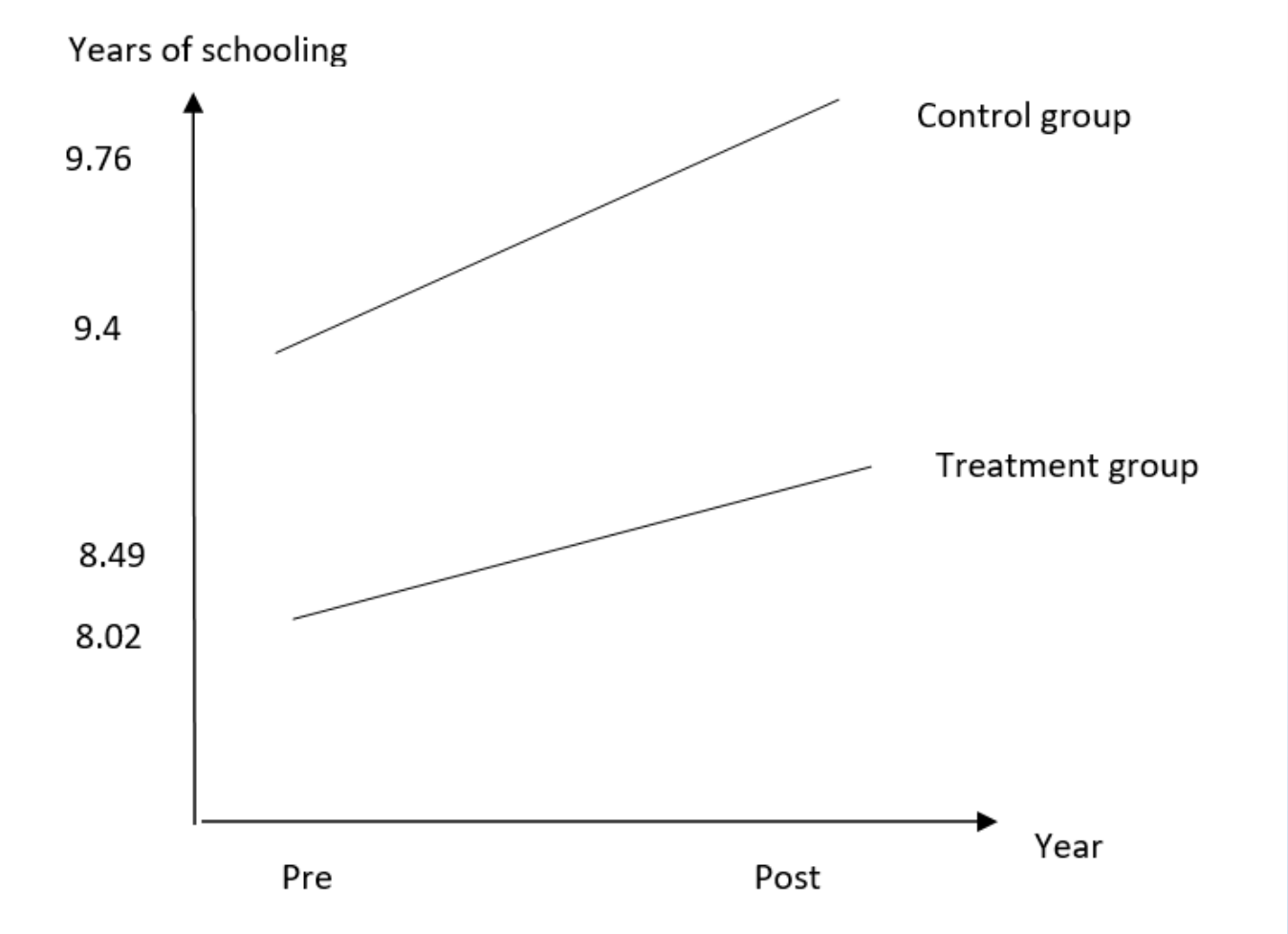
Short Answer

What's the problem with Treatment vs. comparison analysis?

Treatment and comparison analysis



The return to education in Indonesia

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- Treatment vs. comparison analysis does not control for selection bias
- More schools were built in those areas that were initially lagging behind – poorer, more remote, less developed communities

The return to education in Indonesia

- In practice, the DD estimator is:

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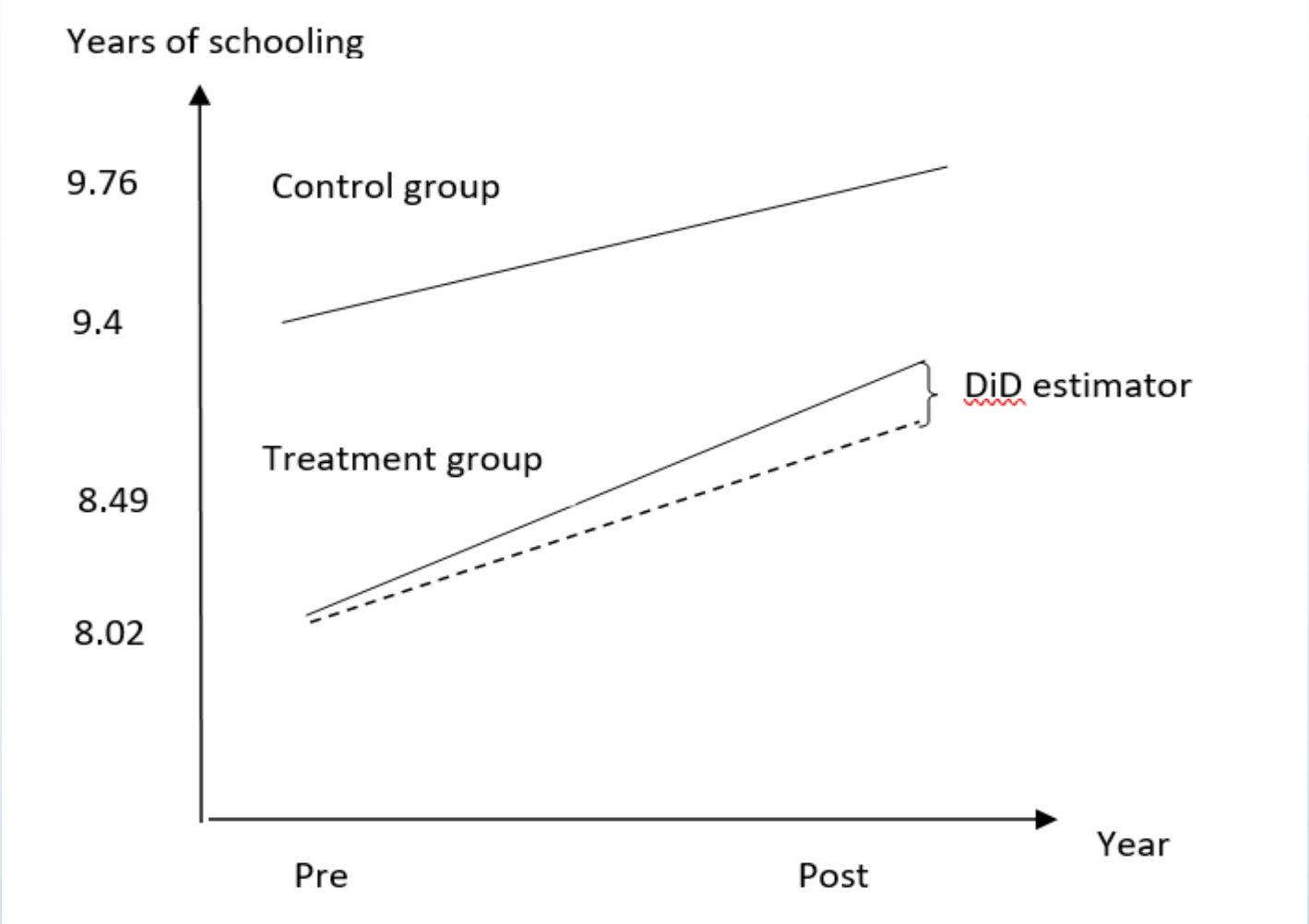
- DD estimator compares the change in years of schooling (i.e. the pre vs. post estimate) in treatment vs control areas
- Program areas increased faster than comparison areas

Difference in differences



The return to education in Indonesia

- Total education attainment and adult wages grew faster in areas where more schools were built as apart of the INPRES program
  - Schools caused an increase in education
  - Increases in education caused an increase in wages
- Same DD strategy to look at wages

	Dependent variable: log (wages)		
	Many schools built	Few schools builts	Difference
Under 7 in 1974	6.61	6.73	-0.12
Over 11 in 1974	6.87	7.02	-0.15
Difference	-0.26	-0.29	0.03

- The earnings of the younger cohort living in the high-construction area rose by an additional 3 percent

The return to education in Indonesia

- Each additional 0.11 years of schooling increased earnings by 3 percent
- This implies that each additional year of school increased earnings by 27 percent (0.03/0.11).
- Therefore, the rate of return to schooling in Indonesia seems quite high, justifying the sizable expenditure made by the school construction program

Practice with random assignment

Random Assignment

People are randomly assigned into a group with intervention(treatment group) and a group without intervention (control group) and then get followed up to see how they fare on various outcomes.

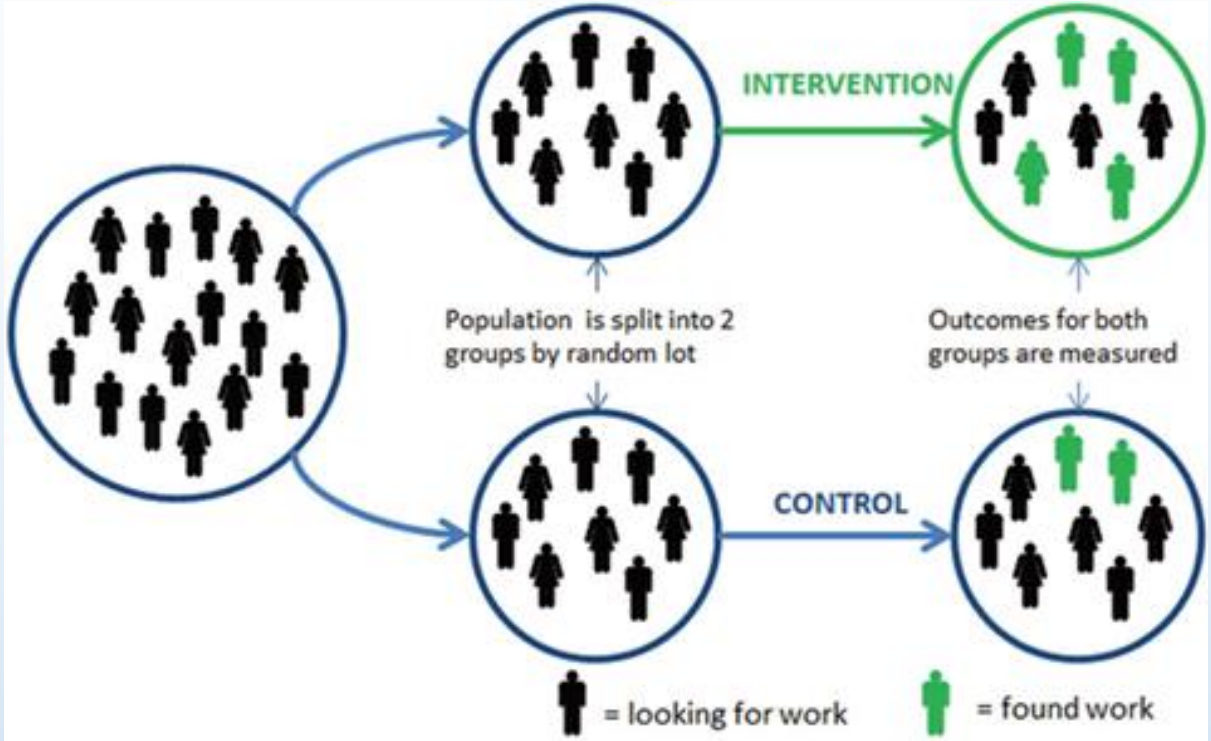


Figure 1. The basic design of a randomised controlled trial (RCT), illustrated with a test of a new 'back to work' programme.

Source: [BIT (publishing.service.gov.uk)]

Practice with Random Assignment

The Ministry of Schools wishes to understand the effect of class sizes on educational attainment. It provides you with a dataset following a group of individuals born in 1972 at seven yearly intervals from birth to age 28. The data contains information on class size at ages 7 and 14, total years of education and earnings at age 28. You run several regressions by OLS aimed at identifying the impact of class size on subsequent economic attainment and get the following:

Dependent variable:	*Earnings*					
	(1)		(2)		(3)	
	Coeff	*t*	Coeff	*t*	Coeff	*t*
Educ	0.22	(5.66)	0.211	(5.36)	0.192	(3.16)
Education squared	-0.0055	(1.70)	-0.0053	(1.65)	-0.006	(2.12)
Class size at 7	-0.063	(1.71)	-0.176	(3.02)		
Class size at 14	-0.18	(2.31)			-0.365	(4.31)
Sampel size	1838		1838		1838	
R-squared	0.314		0.281		0.29	

Note: Number in parentheses are absolute t-values

where Earnings is annual earnings at age 25 in pounds, Education is years of age at completion of education, and Class size at age a is class size at age a in numbers of children per class.

(e) In the 1980s, the US Department for Education implemented a randomized experiment in which several thousands of students were randomly assigned to attend either a small class with an average of 15 students or a regular-sized class with an average of 22 students. The students remained in their randomly assigned classes for a year before returning to their normal class. Data on the students were collected before the experiment started and after it ended until the students reached adulthood (test scores, wages, etc.). In which ways could running a similar experiment in UK schools help the Ministry to assess the effect of class size on later attainment? Be precise about the assumptions that it would allow to validate.

Short Answer

In which ways could running a similar experiment in UK schools help the Ministry to assess the effect of class size on later attainment? Be precise about the assumptions that it would allow to validate.

Answer: Running a randomized experiment whereby randomly chosen group of students would be assigned to a small class while others would be assigned to a large class would allow to identify the causal effect of class size on later attainment. By randomly assigning students to classes of different sizes, we would ensure that class size is uncorrelated with any unobserved factors that might influence later attainment. The two groups of students were randomly picked so the two groups of students should have the same characteristics on average, except for the size of the class they attended. As a result, any difference in later attainment between the two groups could only be due to differences in class size.

Next Week (Reading Week)

- Tutorial 3 assignment submission is due on Wednesday, 18 Feb.

- Empirical group project will be released on 16 Feb, Monday.
  - Group formation guidance at https://moodle.ucl.ac.uk/mod/choicegroup/view.php?id=8102983
  - Ensure to form your group by 26 February 2026 at 2pm
  - Project submission deadline: 16 March.